MATHS MANTHAN
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Subject: Mathematics | **CBSE CLASS 10 – WEEKLY TEST 04** | **Time:** 45 Minutes
Class: Class X (CBSE) | **Chapter 1: Real Numbers** | **MM:** 20

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit theorem/formula declaration, complete prime factorisation or contradiction steps, and boxed final answers with units.
2. In Assertion–Reason questions, truth values of both (A) and (R) as well as whether (R) validly explains (A) are explicitly established.
3. Partial marks are allocated per step in brackets: **[0.5M, 1M, 2M]**. Alternative valid methods receive full credit.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: MCQs & Assertion–Reason

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Q1. If $a = 2^2 \times 3^3 \times 5^4$ and $b = 2^3 \times 3^2 \times 5$, then HCF$(a, b)$ is equal to: [1]

(a) 90 (b) 180 (c) 360 (d) 540

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (b) 180

Given $a = 2^2 \times 3^3 \times 5^4$ and $b = 2^3 \times 3^2 \times 5^1$. We know:

Formula Used: HCF(a, b) = Product of smallest powers of common prime factors

$\text{HCF}(a, b) = 2^{\min(2,3)} \times 3^{\min(3,2)} \times 5^{\min(4,1)} = 2^2 \times 3^2 \times 5^1 = 4 \times 9 \times 5 = \boxed{180}$. [1M]

Q2. The total number of factors of a prime number is: [1]

(a) 1 (b) 2 (c) 0 (d) 3

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (b) 2

We know the definition of a prime number:

Formula Used: A prime $p > 1$ has exactly two distinct positive divisors: 1 and p

Therefore every prime number has $\boxed{2}$ factors. [1M]

Q3. If the HCF of 65 and 117 is expressible in the form $65m - 117$, then the value of m is: [1]

(a) 4 (b) 2 (c) 1 (d) 3

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (b) 2

Prime factorising $65 = 5 \times 13$ and $117 = 3^2 \times 13$. We know:

Formula Used: $\text{HCF}(65, 117) = 13$

Equating $65m - 117 = 13 \implies 65m = 13 + 117 = 130 \implies m = \frac{130}{65} = \boxed{2}$. [1M]

Q4. If p and q are two co-prime natural numbers, then $\text{HCF}(p, q)$ is equal to: [1]

(a) pq (b) $p + q$ (c) 1 (d) 0

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (c) 1

We know the definition of co-prime integers:

Formula Used: Two numbers p, q are co-prime $\iff \text{HCF}(p, q) = 1$

Their only common positive divisor is 1, so $\text{HCF}(p, q) = \boxed{1}$. [1M]

Q5. Assertion (A): $\sqrt{2}$ is an irrational number.
Reason (R): If p is a prime number, then $\sqrt{p}$ is an irrational number. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (A) Both (A) and (R) are true and (R) is the correct explanation of (A).

We know the general theorem on prime roots:

Formula Used: p prime $\implies \sqrt{p}$ is irrational

Reason (R) states this true theorem; applying it to the prime number $p = 2$ proves $\sqrt{2}$ is irrational (Assertion A). Thus both (A) and (R) are true and (R) explains (A). [1M]

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SECTION B: Very Short Answer Type — MATHS MANTHAN | (1 × 2 = 2 Marks)

Q6. Given that HCF$(306, 657) = 9$, find LCM$(306, 657)$. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Given: $a = 306$, $b = 657$, and HCF$(306, 657) = 9$. We know the product identity:

Formula Used: $\text{HCF}(a,b) \times \text{LCM}(a,b) = a \times b \implies \text{LCM}(a,b) = \dfrac{a \times b}{\text{HCF}(a,b)}$

Substituting and simplifying cleanly before multiplying:

$$\text{LCM}(306, 657) = \frac{306 \times 657}{9} = 34 \times 657. \qquad [1\text{M}]$$

Multiplying step by step: $34 \times 657 = (30 + 4) \times 657 = 19710 + 2628 = \boxed{22338}$. [1M]

SECTION C: Short Answer Type — MATHS MANTHAN | (3 × 3 = 9 Marks)

Q7. Find the HCF and LCM of 288, 360, and 384 by the prime factorisation method. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Prime factorising each integer into prime-power form:

$$288 = 32 \times 9 = 2^5 \times 3^2, \qquad 360 = 8 \times 9 \times 5 = 2^3 \times 3^2 \times 5^1, \qquad 384 = 128 \times 3 = 2^7 \times 3^1. \qquad [1\text{M}]$$

We know the HCF and LCM prime-exponent rules:

Formula Used: $\text{HCF} = \prod p_i^{\min\ \text{exp}}, \qquad \text{LCM} = \prod p_i^{\max\ \text{exp}}$

HCF: Common primes are 2 and 3, with smallest powers 2^3 and 3^1:

$$\text{HCF}(288, 360, 384) = 2^3 \times 3^1 = 8 \times 3 = \boxed{24}. \qquad [1\text{M}]$$

LCM: All primes $2, 3, 5$ with highest powers $2^7, 3^2, 5^1$:

$$\text{LCM}(288, 360, 384) = 2^7 \times 3^2 \times 5 = 128 \times 9 \times 5 = 128 \times 45 = \boxed{5760}. \qquad [1\text{M}]$$

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Q8. An army contingent of 616 members is to march behind an army band of 32 members in a parade. The two groups are to march in the same number of columns. What is the maximum number of columns in which they can march? **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Given: Contingent size = 616, Band size = 32. Maximum equal columns dividing both exactly equals HCF(616, 32). We know:

Formula Used: Maximum equal columns = HCF(616, 32)

Prime factorising 616 and 32:

$$616 = 8 \times 77 = 2^3 \times 7 \times 11, \qquad 32 = 2^5. \qquad \textbf{[1.5M]}$$

(Equivalently, Euclid's division: $616 = 32 \times 19 + 8$, $32 = 8 \times 4 + 0$.) Taking common prime factor with smallest exponent:

$$\text{HCF}(616, 32) = 2^3 = \boxed{8 \text{ columns}}. \qquad \textbf{[1.5M]}$$

Q9. A sweet seller has 420 kaju burfis and 130 badam burfis. She wants to stack them in such a way that each stack has the same number, and they take up the least area of the tray. What is the number of burfis that can be placed in each stack for this purpose? **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

To occupy the least tray area, each stack must hold the maximum equal number of burfis, which is HCF(420, 130). We know:

Formula Used: Optimal stack size = HCF(420, 130)

Prime factorising 420 and 130:

$$420 = 10 \times 42 = 2^2 \times 3 \times 5 \times 7, \qquad 130 = 2 \times 5 \times 13. \qquad \textbf{[1.5M]}$$

Common primes are 2 and 5, each with least power 1 (or by Euclid: $420 = 130 \times 3 + 30$, $130 = 30 \times 4 + 10$, $30 = 10 \times 3 + 0$):

$$\text{HCF}(420, 130) = 2 \times 5 = \boxed{10 \text{ burfis per stack}}. \qquad \textbf{[1.5M]}$$

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Q10. CASE STUDY 04: INDUSTRIAL ELECTRONIC ALARM CHIMES WARNING SIRENS [4 Marks]

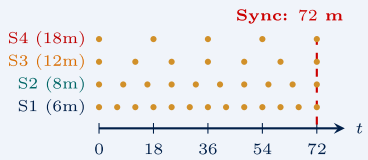
A modern automated production factory employs a synchronized auditory signal system to coordinate manufacturing shifts, machine safety checks, cooling inspections, and supervisor walk-throughs. Four distinct sirens sound automatically at regular intervals of 6 minutes, 8 minutes, 12 minutes, and 18 minutes respectively. All four sirens are synchronized and sound simultaneously at 8:00 AM.

(i) Find the least common multiple (LCM) of $6, 8, 12$, and 18 using the prime factorisation method. [2]

(ii) At what exact clock time will all four sirens next sound simultaneously? [1]

(iii) How many times will all four sirens sound simultaneously during a 12-hour factory shift (including the initial toll at 8:00 AM)? [1]

▶ CASE-BASED SOLUTION & MARKING SCHEME: [Total: 4 Marks]



Synchronization Timeline: All four sirens $(6, 8, 12, 18$ min) realign every 72 min (1 hr 12 min). Starting at 8:00 AM, the 10 subsequent chimes occur up to 8:00 PM (720 min).

We know the synchronization time formula via LCM:

Formula Used: Synchronized interval $= \text{LCM}(6, 8, 12, 18)$, Tolls (inclusive) $= \dfrac{T_{\text{total}}}{T_{\text{sync}}} + 1$

(i) LCM of $6, 8, 12, 18$:

$$6 = 2^1 \times 3^1, \quad 8 = 2^3, \quad 12 = 2^2 \times 3^1, \quad 18 = 2^1 \times 3^2.$$

Taking highest powers of primes 2 and 3:

$$\text{LCM}(6, 8, 12, 18) = 2^3 \times 3^2 = 8 \times 9 = \boxed{72 \text{ minutes}}. \qquad [2\text{M}]$$

(ii) Next simultaneous chime time: Adding 72 min = 1 hr 12 min to 8:00 AM gives $\boxed{9{:}12 \text{ AM}}$. [1M]

(iii) Total simultaneous chimes in 12 hours: Total shift $= 12 \times 60 = 720$ minutes. Since $720 \div 72 = 10$ exact intervals after 8:00 AM, total tolls including the initial toll at 8:00 AM (and 10th interval ending at 8:00 PM) is $10 + 1 = \boxed{11 \text{ times}}$. [1M]

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★